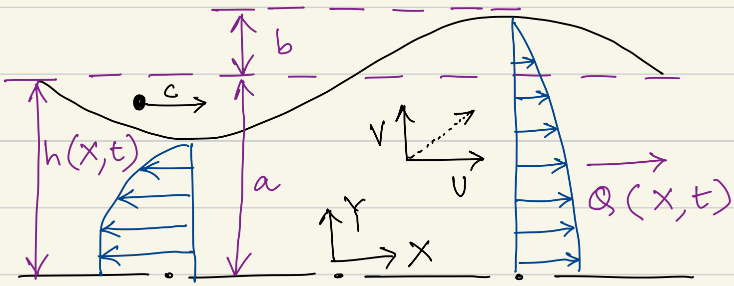
Peristaltic Pumping with long wavelengths at low Reynolds number – Shapiro, Jaffrin & Weinberg, 1969

Pumping for an infinite train of peristaltic waves in a 2D channel.



The laboratory reference frame (Eulerian)

The wave number, a/λ, $a \rightarrow$ mean half width of channel

$\lambda \rightarrow$ wavelength

The amplitude ratio, $\phi = b/a$, $b \rightarrow$ The half amplitude of peristaltic wave.

The Reynolds Number, $R = \dfrac{ac}{\nu}\dfrac{a}{\lambda}$, $c \rightarrow$ is wave speed.

Dimensionless time mean flow, $\Theta = \dfrac{\bar{Q}}{bc}$, $\bar{Q} \rightarrow$ time-mean flow observed in the laboratory frame.

The effect of inertia

The leading viscous term in the equation of motion,

$$\mu \frac{\partial^2 u}{\partial y^2} \sim \mu \frac{c}{a^2} \quad , \text{and} \quad \frac{\partial u}{\partial x} \sim \frac{c}{\lambda}$$

The typical inertia term scales as , $\rho u \frac{\partial u}{\partial x} \sim \rho \frac{c^2}{\lambda}$

Hence, the ratio of inertia to viscous forces is,

Reynolds num: $R \sim \frac{\rho c^2}{\lambda} \cdot \frac{a^2}{\mu c} \sim \left(\frac{c\,a}{\nu}\right)\left(\frac{a}{\lambda}\right)$, where $\nu = \frac{\mu}{\rho}$

The infinite sinusoidal wave-train on top and bottom wall,

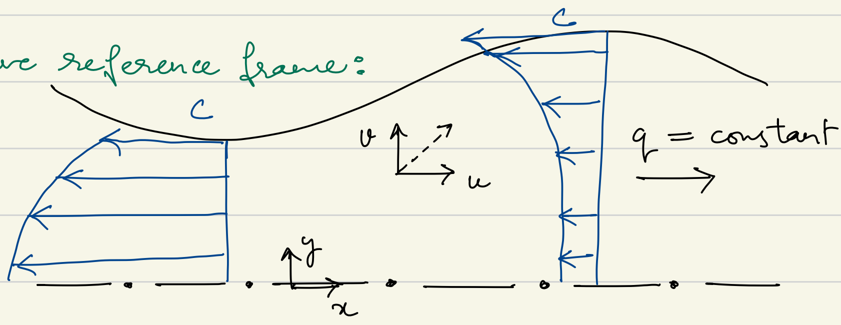
$$H = 1 + \phi \sin\left(2\pi\left(\xi - \tau\right)\right) \quad \Big\} \text{ In the lab frame}$$

$H = \frac{h}{a} \rightarrow$ is the dimensionless wall distance coordinate.

$\xi = \frac{x}{\lambda} \rightarrow$ the dimensionless axial coordinate

$\tau = \frac{ct}{\lambda} \rightarrow$ the dimensionless time

The wave reference frame:



The transformation between the two frames

$$x = X - ct \qquad u(x,y) = U(X-ct, Y) - c$$
$$y = Y \qquad v(x,y) = V(X-ct, Y)$$

The Navier Stokes equation, $\rho\left(\frac{\partial \vec{u}}{\partial t} + \vec{u}\cdot\vec{\nabla}\vec{u}\right) = -\vec{\nabla}p + \mu\nabla^2\vec{u}$

The lubrication approximation in Shapiro et al.

$u \sim c$, $x \sim \lambda$, $y \sim a$ and $\frac{\partial u}{\partial y} \sim \frac{c}{a}$

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \frac{\partial v}{\partial y} \sim \frac{\partial u}{\partial x} \sim \frac{c}{\lambda} \Rightarrow v \sim \frac{ca}{\lambda}$$

$$\frac{\partial v}{\partial x} \sim \frac{1}{\lambda}\frac{ca}{\lambda} \sim \frac{ca}{\lambda^2}, \quad \frac{\partial u}{\partial y} \sim \frac{c}{a}, \quad \frac{\partial u}{\partial x} \sim \frac{c}{\lambda}, \quad \frac{\partial v}{\partial y} \sim \frac{c}{\lambda}$$

We can define the lubrication smallness param : $\epsilon = \frac{a}{\lambda}$
and find the scaling w.r.t. $\frac{\partial u}{\partial y}$

$$\Rightarrow \frac{\partial v}{\partial x} \sim \epsilon^2 \frac{\partial u}{\partial y} \; ; \; \frac{\partial v}{\partial y} \sim \epsilon \frac{\partial u}{\partial y} \; ; \; \frac{\partial u}{\partial x} \sim \epsilon \frac{\partial u}{\partial y}$$

$$\left(\frac{\partial u}{\partial y}\right) \sim O(1) \Rightarrow \frac{\partial v}{\partial x} \sim O(\epsilon^2) \;\&\; \frac{\partial v}{\partial y} \sim \frac{\partial u}{\partial x} \sim O(\epsilon)$$

Non dimensionalizing NS eq: $u^* c = u$, $t^* \frac{\lambda}{c} = t$, $x^* \lambda = x$, $y^* a = y$

$v^* \frac{ca}{\lambda} = v$, $p^* \frac{\mu c}{a^2} \lambda = p$ $\rightarrow$ The pressure scaling chosen to balance leading term $\mu \partial^2 u / \partial y^2$

$$\Rightarrow \rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = - \frac{\partial p}{\partial x} + \mu \frac{\partial^2 u}{\partial x^2} + \mu \frac{\partial^2 u}{\partial y^2}$$

$$\rho \left(\frac{c^{\cancel{2}}}{\lambda} \frac{\partial u^*}{\partial t^*} + \frac{u^* c^{\cancel{2}}}{\lambda} \frac{\partial u^*}{\partial x^*} + v^* \frac{c^{\cancel{2}} \cancel{a}}{\lambda \cancel{a}} \frac{\partial u^*}{\partial y^*} \right) = - \frac{\mu \cancel{c}}{a^2} \frac{\partial p^*}{\partial x^*} + \mu \frac{\cancel{c}}{\lambda^2} \frac{\partial^2 u^*}{\partial x^{*2}}$$

$$+ \mu \frac{\cancel{c}}{a^2} \frac{\partial^2 u^*}{\partial y^{*2}}$$

$$\rho \frac{c}{\lambda} \left(\frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} \right) = \frac{\mu}{a^2} \left(- \frac{\partial p^*}{\partial x^*} + \frac{a^2}{\lambda^2} \frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}} \right)$$

$$\Rightarrow \left(\underbrace{\frac{ca}{\nu}}_{R} \cdot \frac{a}{\lambda} \right) \left(\frac{\partial u^*}{\partial t^*} + u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} \right) = \left(- \frac{\partial p^*}{\partial x^*} + \left(\frac{a}{\lambda} \right)^2 \frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}} \right)$$

, we can use Reynolds number, $R \rightarrow 0$ and $\frac{a}{\lambda} \rightarrow 0$

$$\Rightarrow \quad \frac{\partial p^*}{\partial x^*} = \frac{\partial^2 u^*}{\partial y^{*2}}$$

$\rightarrow$ The x-momentum equation to solve.

$$\text{And, } \rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) = - \frac{\partial p}{\partial y} + \mu \frac{\partial^2 v}{\partial x^2} + \mu \frac{\partial^2 v}{\partial y^2}$$

$$\Rightarrow \rho \left(\frac{ca}{\lambda} \frac{\cancel{c}}{\lambda} \frac{\partial v^*}{\partial t^*} + \frac{c^{\cancel{2}} a}{\lambda^2} u^* \frac{\partial v^*}{\partial x^*} + \frac{c^{\cancel{2}} a^{\cancel{2}}}{\lambda^2} \frac{1}{\cancel{a}} v^* \frac{\partial v^*}{\partial y^*} \right) = - \frac{\mu \cancel{c} \lambda}{a^2 \cdot a} \frac{\partial p^*}{\partial y^*}$$

$$+ \frac{\mu}{\lambda^2} \frac{\cancel{c} a}{\lambda} \frac{\partial^2 v^*}{\partial x^{*2}} + \frac{\mu}{a^2} \frac{\cancel{c} a}{\lambda} \frac{\partial^2 v^*}{\partial y^{*2}}$$

$$\Rightarrow \frac{\rho c a}{\lambda^2} \times \frac{a^3}{\mu \lambda} \left(\frac{\partial v^*}{\partial t^*} + u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} \right) = -\frac{\partial p^*}{\partial y^*} + \frac{a^3}{\cancel{\mu}\lambda} \frac{\cancel{\mu} a}{\lambda^3} \frac{\partial^2 v^*}{\partial x^{*2}}$$

$$\left(\frac{ca}{\nu} \times \frac{a}{\lambda} \right) \frac{a^2}{\lambda^2}$$

$$+ \frac{a^3}{\cancel{\mu}\lambda} \frac{\cancel{\mu} a}{a^2 \lambda} \frac{\partial^2 v^*}{\partial y^{*2}}$$

$$R \times \epsilon^2 \left(\qquad \right) = -\frac{\partial p^*}{\partial y^*} + \epsilon^4 \frac{\partial^2 v^*}{\partial x^{*2}} + \epsilon^2 \frac{\partial^2 v^*}{\partial y^{*2}}$$

as $R \rightarrow 0$, we have LHS $\rightarrow 0$

and for lubrication assumption, $\epsilon \rightarrow 0 \Rightarrow \dfrac{\partial p^*}{\partial y^*} = 0$

The y-mom equation to solve

To obtain the solution:

$$U = u + c \Rightarrow u = U - c$$

and $\quad x = X - ct$ and $y = Y$

$$H = 1 + \phi \sin\left[2\pi\left(\xi - \tau\right)\right] ; \quad \xi = \frac{X}{\lambda}, \quad H = \frac{h}{a}$$

$$\tau = \frac{ct}{\lambda}$$

$$\Rightarrow h = a + \phi a \sin\left[\frac{2\pi}{\lambda}\left(X - ct\right)\right]$$

$$h = a + b \sin\left(2\pi x / \lambda\right) \longrightarrow$$

λ is the wavelength and $\phi = \dfrac{b}{a}$ is the wall deflection ratio.

The momentum equation in the wave frame.

$$\frac{\partial p}{\partial x} = \mu \frac{\partial^2 u}{\partial y^2}$$

The boundary conditions, $u=-c$ at $y=h(x) \rightarrow$ no-slip.

and symmetry boundary as $\frac{\partial u}{\partial y}=0$ at $y=0$

$$\frac{\partial p}{\partial y}=0 \Rightarrow \int \frac{\partial p}{\partial y}\,dy = \int 0\,dy \Rightarrow \underbrace{p = p(x)}_{\text{pressure is a function only of } x.} + K_0$$

Now, $$\frac{\partial p}{\partial x} = \mu \frac{\partial^2 u}{\partial y^2} \Rightarrow \int \frac{\partial p}{\partial x}\,dy = \mu \int \frac{\partial^2 u}{\partial y^2}\,dy$$

$$\frac{\partial p}{\partial x} y + K_1 = \mu \frac{\partial u}{\partial y}$$ $\rightarrow$ K_1 is the lumped integration constant

$$\Rightarrow \int \frac{\partial u}{\partial y}\,dy = \int \left(\frac{y}{\mu}\frac{\partial p}{\partial x} + K_1\right) dy$$

$$\Rightarrow u = \frac{y^2}{2\mu}\frac{\partial p}{\partial x} + K_1 y + K_2$$

using B.C. $\frac{\partial u}{\partial y}=0$ at $y=0 \Rightarrow K_1 = 0$ and

$u=-c$ at $y=h(x) \Rightarrow -c = \frac{h^2(x)}{2\mu}\frac{\partial p}{\partial x} + K_2$ $\left\}\right.$ we can just use $\frac{\partial p}{\partial x} = \frac{dp}{dx}$

$$\Rightarrow u = \frac{y^2}{2\mu}\frac{dp}{dx} - c - \frac{h^2(x)}{2\mu}\frac{dp}{dx} \Rightarrow u = \frac{1}{2\mu}\frac{dp}{dx}\left(y^2 - h^2(x)\right) - c$$

In Shapiro et al. 1969, this is written as,

$$\frac{u}{c} = -1 - \frac{a^2}{2\mu c}\frac{dp}{dx}\left[\left(\frac{h(x)}{a}\right)^2 - \left(\frac{y}{a}\right)^2\right]$$

The flux in the half channel is,

$$q = \int_0^h u\,dy = \int_0^h \left[\frac{1}{2\mu}\frac{dp}{dx}\left(y^2 - h^2(x)\right) - c\right] dy$$

$$= \frac{1}{2\mu}\frac{dp}{dx}\left(\frac{y^3}{3} - h^2(x)\,y\right)\Big|_0^h - ch + 0$$

$$= \frac{1}{2\mu}\frac{dp}{dx}\left[\frac{h^3}{3} - h^3 - 0 + 0\right] - ch$$

$$= \frac{1}{2\mu}\frac{dp}{dx}\left[\frac{-2h^3}{3}\right] - ch = \frac{-1}{\mu}\frac{dp}{dx}\frac{h^3(x)}{3} - ch$$

$$\Rightarrow \frac{q}{ac} = -\frac{h}{a} - \frac{h^3}{3\mu c a}\frac{dp}{dx}$$ in Shapiro et al (1969)

$$\Rightarrow \frac{dp}{dx} = \frac{-3\mu}{h^3(x)}\left(q + c\,h(x)\right)$$

Now, $Q = \int_0^h U\,dY = \int_0^h (u+c)\,dy = q + cy\Big|_0^h$

$$Q = q + c\,h(x)$$

Now, we take the steady flow in the laboratory frame.

$$\bar{Q} = \frac{1}{T}\int_0^T Q\,dt = \frac{1}{T}\int_0^T q\,dt + \frac{c}{T}\int_0^T h(x)\,dt$$

$x = X - ct$

$dx = dX - c\,dt$

at constant X, $dX = 0$

$\Rightarrow dx = -c\,dt$

change of limits,

$t = 0$, $x = X$ and $t = T$, $x = X - cT = X - \frac{\lambda}{\cancel{T}}\cancel{T}$

$x = X - \lambda$

$$\Rightarrow \bar{Q} = \bar{q} + \frac{c}{T}\int_X^{X-\lambda}\left[a + a\phi \sin(2\pi x/\lambda)\right]\left(\frac{dx}{-c}\right)$$

$c = \frac{\lambda}{T}$

$\frac{c}{\lambda} = \frac{1}{T}$

$$= \underbrace{q}_{\bar{q} = q} - \frac{c}{\lambda}\left[ax + a\phi\,\frac{-\cos(2\pi x/\lambda)}{2\pi/\lambda}\right]_X^{X-\lambda}$$

$$= q - \frac{c}{\lambda}\left[a(X-\lambda-X) - a\phi\,\underbrace{\cos(-2\pi)\frac{\lambda}{2\pi}}_{0}\right]$$

$$\bar{Q} = q + ca \Rightarrow q = \bar{Q} - ca$$

Writing the solution in the lab frame.

$$u = \frac{1}{2\mu}\frac{dp}{dx}\left(y^2 - h^2(x)\right) - c = \frac{1}{2\cancel{\mu}}\,\frac{-3\cancel{\mu}}{h^3(x)}\left(q + ch(x)\right) - c$$

$$\Rightarrow U = u + c = -\frac{3}{2}\,\frac{\left(\bar{Q} - ca + ch(x)\right)}{h^3(x)}\left[y^2 - h^2(x)\right]$$

The final solution is therefore given not just by $h(x)$ but also by either dp/dx or by $\bar{Q}$, which are given by the channel inlet-outlet boundary condition.

We consider two special cases:

The "free pumping" condition — This happens when we have periodic boundaries with zero pressure drop across wavelength λ

The pressure rise per wavelength,

$$\Delta p_\lambda = \int_0^\lambda \left(\frac{dp}{dx}\right) dx = \int_0^\lambda -\frac{3\mu}{h^3(x)}(q + ch)\, dx$$

After substituting for $h(x)$ and q from the previous relation the integration gives,

$$\frac{a^2}{\mu c \lambda} \Delta p_\lambda = \frac{3}{2} \frac{\phi^2}{(1-\phi^2)^{5/2}} \left[3 - \frac{2+\phi^2}{\phi} \theta \right]$$

} Δp_λ gives the pressure rise per wavelength λ

and $\theta = \dfrac{\bar{Q}}{bc}$ is

for "free pumping" condition $\Delta p_\lambda = 0$

$$\Rightarrow \quad \theta = \frac{3\phi}{(2+\phi^2)} \quad \Rightarrow \quad \frac{\bar{Q}}{bc} = \frac{3\phi}{(2+\phi^2)}$$

$$\Rightarrow \boxed{\bar{Q} = \frac{3ac\phi^2}{(2+\phi^2)}}$$

} This is the "free pumping" flux for wavespeed c.

Another condition would be when $\bar{Q} = 0 \Rightarrow \theta = \frac{\bar{Q}}{bc} = 0$

Taking,
$$\frac{a^2}{\mu c \lambda} \Delta p_\lambda \Big|_{\theta=0} = \frac{3}{2} \frac{\phi^2}{(1-\phi^2)^{5/2}} \left[3 - \cancel{\frac{2+\phi^2}{\phi} \theta} \right]$$

0

$$\boxed{\frac{a^2}{\mu c \lambda} \Delta p_\lambda \Big|_{\theta=0} = \frac{9}{2} \frac{\phi^2}{(1-\phi^2)^{5/2}}}$$